

Machine Learning for social sciences

Machine Learning: Session 2

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Regression is one of the most widely used techniques in data science, economics and social sciences.

Motivating Example

Scenario:

- You're analyzing how students perform based on different factors.
- Data collected:
 - Time spent studying in hours
 - Time spent sleeping in hours
 - Attendance at the classes
 - Exam score

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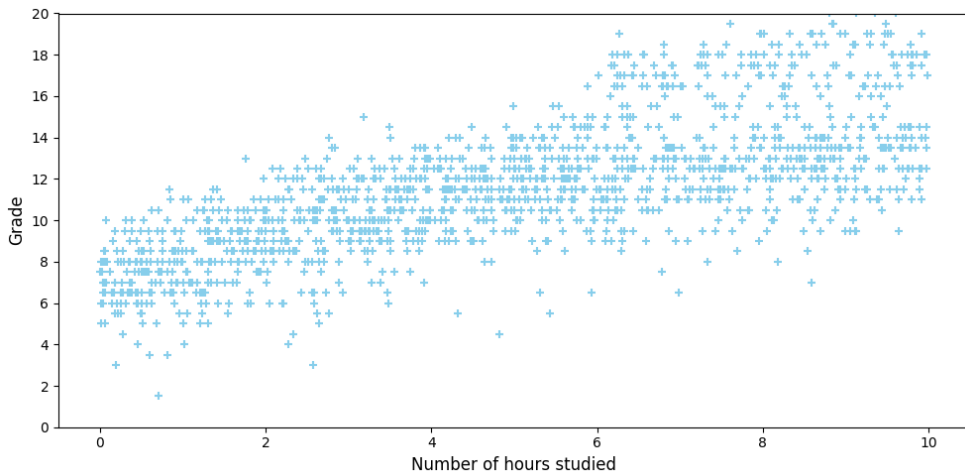
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Key Question:

If a student studies for 5 hours, attends all classes but has had a bad night's sleep before the exam, what score can we expect?

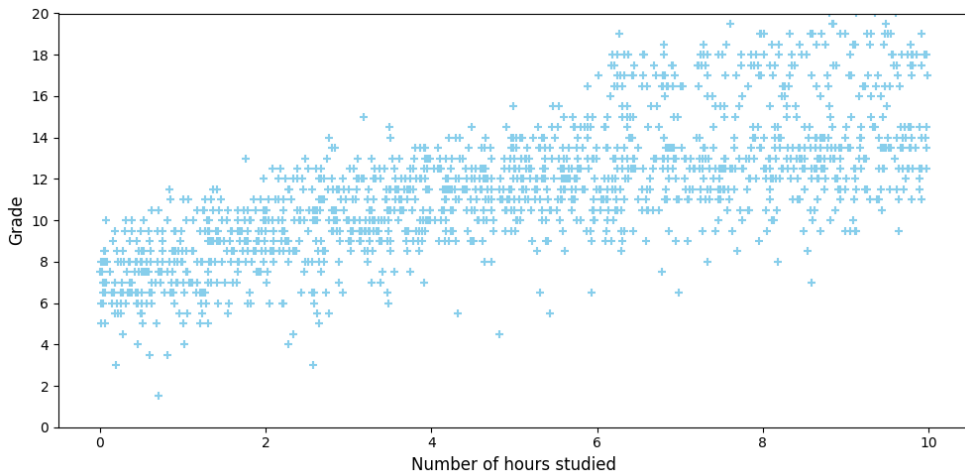
Visualizing the data

Partial representation of the data we will work with:



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Is the relationship linear? Nonlinear? Binary?

What's Coming Up

We'll examine 3 types of regression using our example:

- ① **Linear Regression** – Predicting exact score.
- ② **Polynomial Regression** – Modeling nonlinear trends.
- ③ **Logistic Regression** – Predicting pass/fail outcomes.

Same data, different modeling goals.

Linear regression

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The method tries to model the relationship between your features $[x_1; x_2; \dots; x_n]$ and your desired output value y as a linear equation:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

With β s the weights given to each factor by **fitting** your equation to the training data, and $\hat{y}$ your predicted output.

Side-note: Model training / fitting

Goal: Find the best-fitting line that minimizes the difference between the predicted and actual values.

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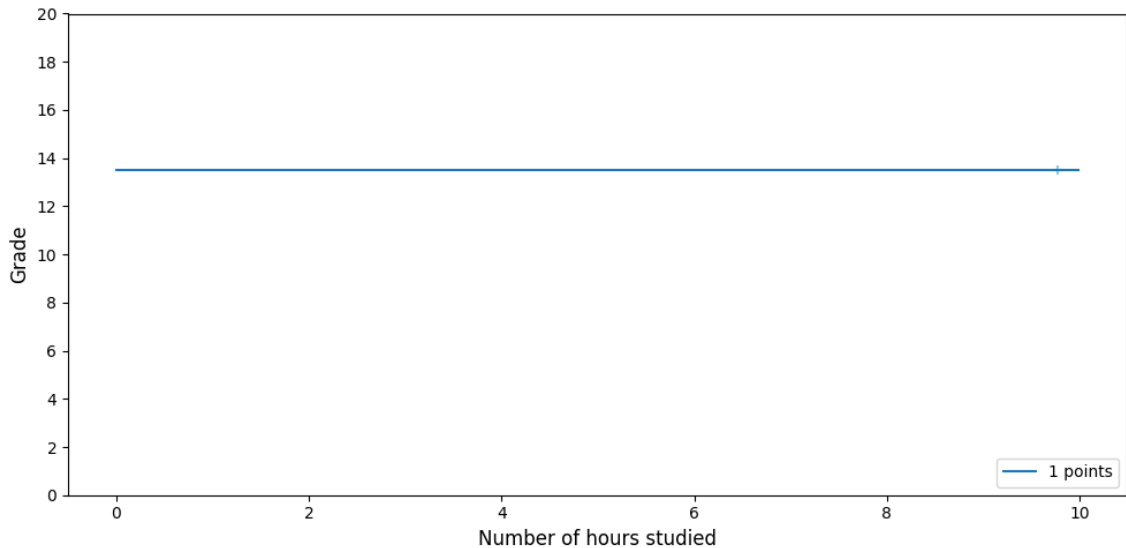
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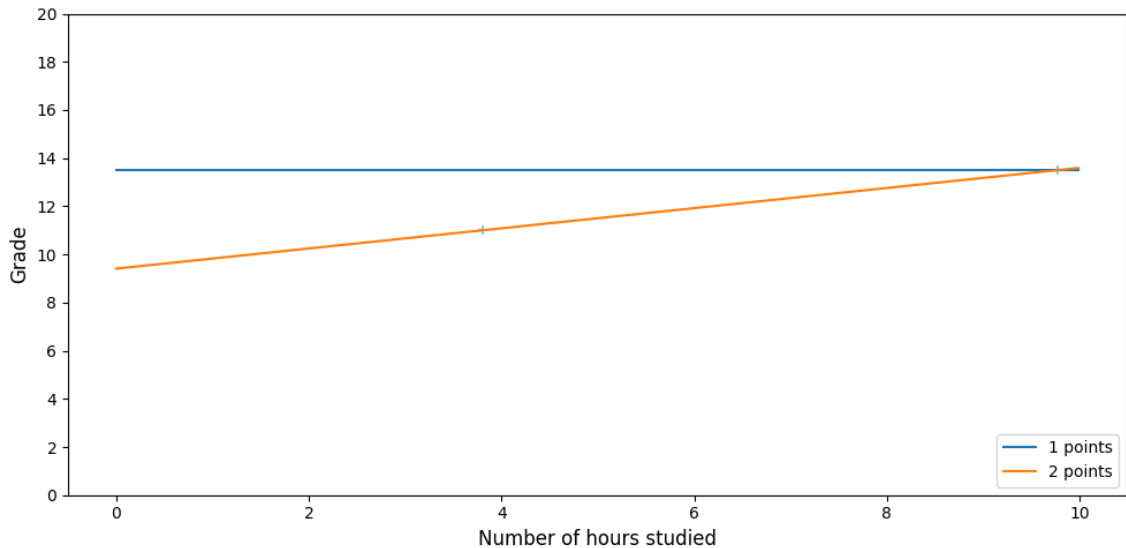
Solution: Choose β_0 and β_1 that minimize the squared error.

This class does not go into the mathematical details of how to compute your β s, we will have a Python library do it for us. But if you are interested, searching for “gradient descent” is a good first step.

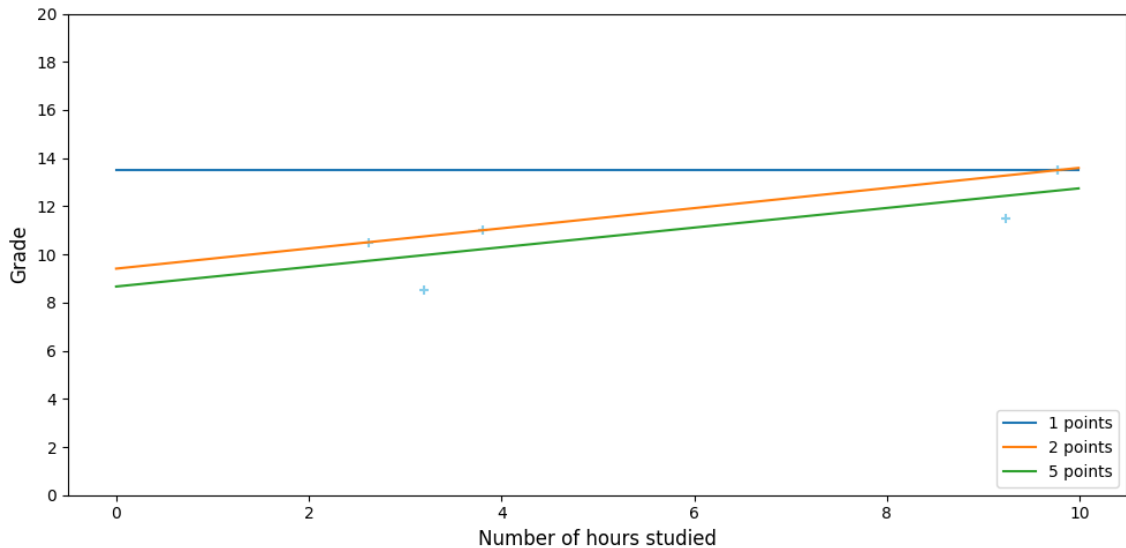
Linear regression in practice



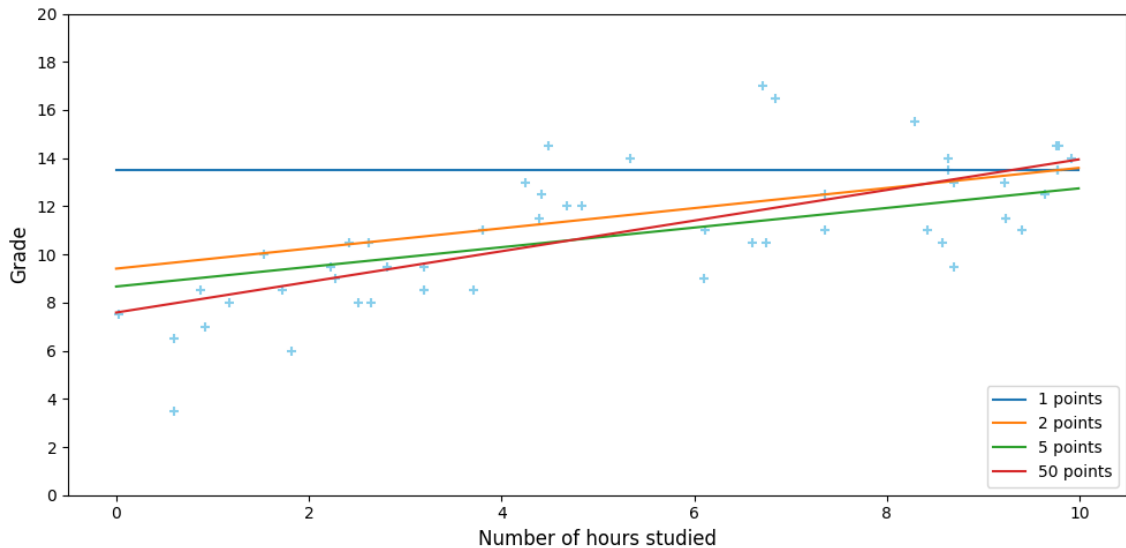
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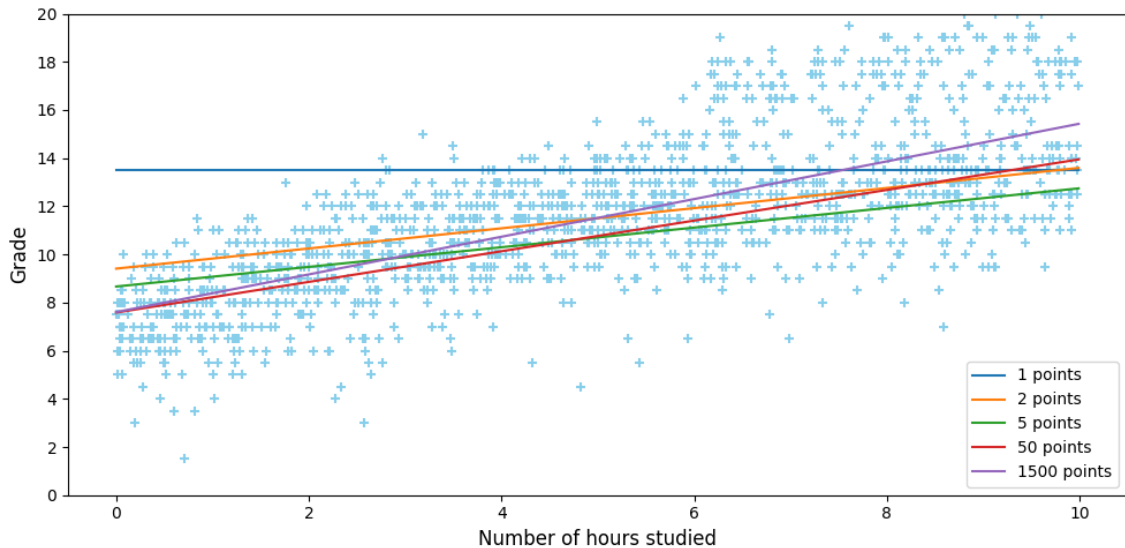
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Polynomial regressions and addition of **interaction terms** allow us to model these more complex relationships.

Polynomial regression

Idea: Extend linear regression by adding powers of the predictor variable(s) to model nonlinear relationships.

General Form:

$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \cdots + \beta_n x^n$$

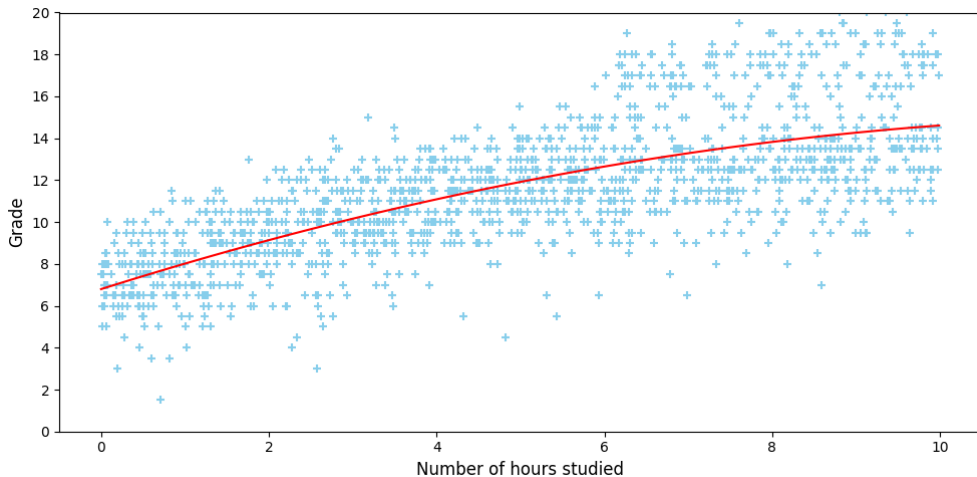
Key Points:

- Still a *linear model* in terms of the parameters β .
- Captures curves and nonlinear patterns.
- Risk of *overfitting* if degree is too high.

Polynomial regression

Example: Predict exam score from hours studied:

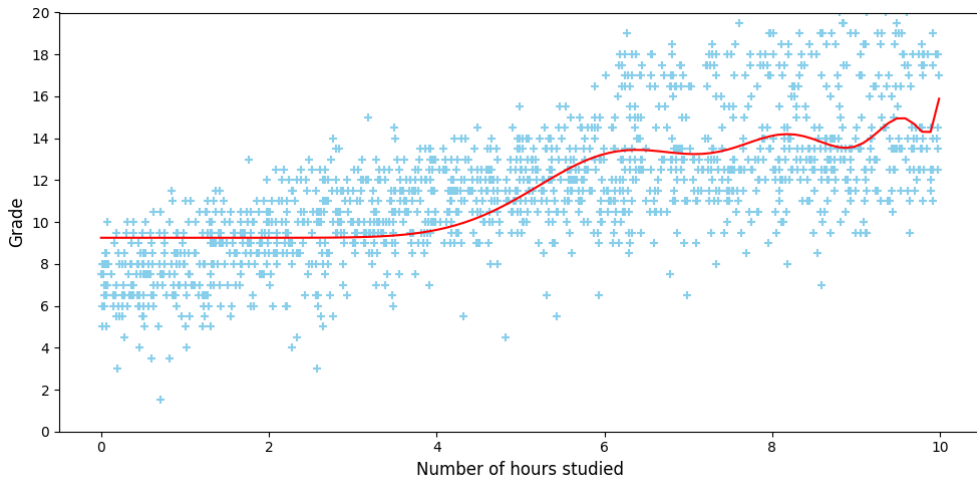
$$\text{Score} = \beta_0 + \beta_1(\text{Hours}) + \beta_2(\text{Hours})^2$$



Polynomial regression

Overfitting example by increasing the degree of the regression:

$$\text{Score} = \beta_0 + \beta_1(\text{Hours}) + \beta_2(\text{Hours})^2 + \dots + \beta_{20}(\text{Hours})^{20}$$



What are interaction terms?

Interaction terms capture the effect of two (or more) variables acting together by adding a product term. For example:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 (x_1 \times x_2)$$

Interpretation: The effect of x_1 depends on the level of x_2 , and vice versa.

Why are they interesting?

- Real-world relationships are rarely additive.
- Capture *conditional effects* between features.
- Reveal synergistic or moderating effects.

Interactions allow more flexible modeling:

- Example: Hours studied (x_1) and sleep quality (x_2) both affect exam score.
- Their combined effect may be larger (or smaller) than the sum of their individual effects.

Shortcomings:

- **Interpretability:** Interaction coefficients can be difficult to explain.
- **Complexity:** The number of possible interactions grows quickly with the number of predictors.
- **Overfitting:** Including unnecessary interactions can harm generalization.
- **Multicollinearity:** Interaction terms often correlate with main effects.

So far: we specified polynomial terms and interaction terms *by hand* to capture nonlinearity and interactions.

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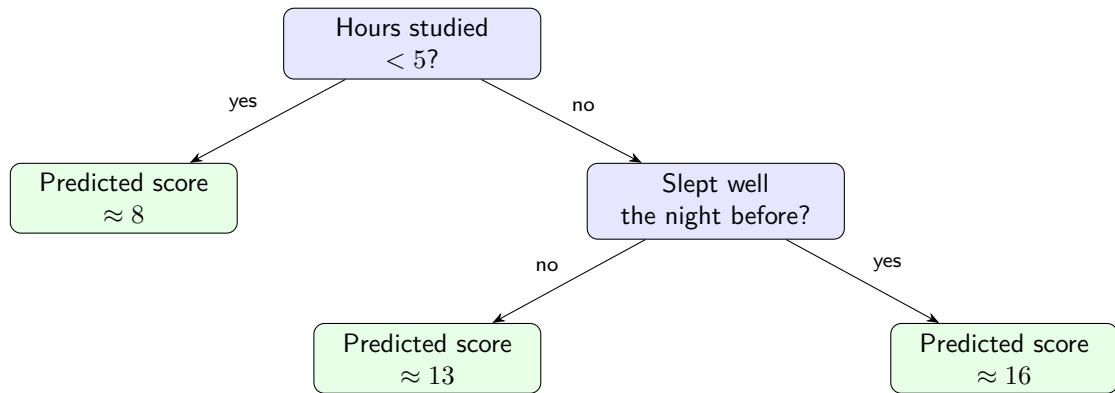
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What if the model could discover these patterns on its own, directly from the data?

This is exactly what **tree-based methods** do – starting with the **decision tree**.

Decision trees

Idea: Repeatedly split the data based on feature values, until each group (a *leaf*) is used to make a prediction.



Decision trees: how they are built

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- ③ Repeat the process recursively on each resulting group.
- ④ Stop once a group is small enough, or pure enough: it becomes a **leaf**, predicting the average outcome of that group.

Strengths:

- Captures **nonlinearities** and **interactions** automatically – no manual x^2 or $x_1 \times x_2$ terms needed.
- Easy to interpret: just follow the branches.

Decision trees: strengths and limits

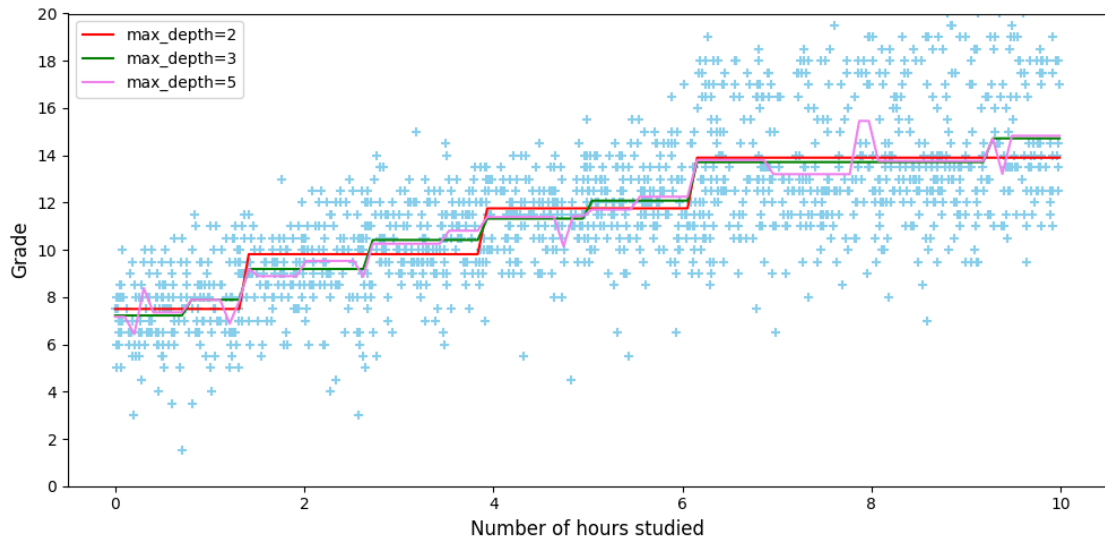
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Limits:

- A single tree is **unstable**: a small change in the training data can produce a very different tree.
- Prone to **overfitting**, especially when allowed to grow deep.

Predictions using a decision tree



From one tree to a forest

Idea: Instead of trusting a single, unstable tree, train **many** trees and combine their predictions.

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This is called **ensemble learning**: combining multiple models usually performs better, and more reliably, than any single one on its own.

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Step 1 – Bootstrap: Create many random samples of the training data, drawn *with replacement*.

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This is the core idea behind **Random Forest**.

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Random Forest adds one more trick: at each split, only a **random subset of features** is considered as a candidate.

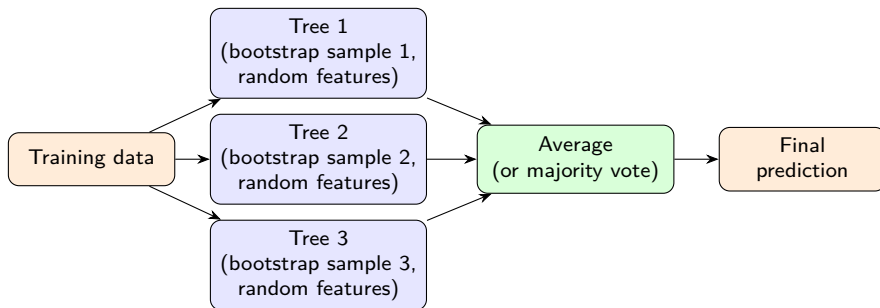
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Effect: trees become more different from each other, so averaging them reduces variance more effectively.

Random forest: putting it all together



(In practice, a random forest typically trains hundreds of trees, not just three.)

Feature importance

Random forests offer a natural way to answer: **which features matter most?**

Feature importance

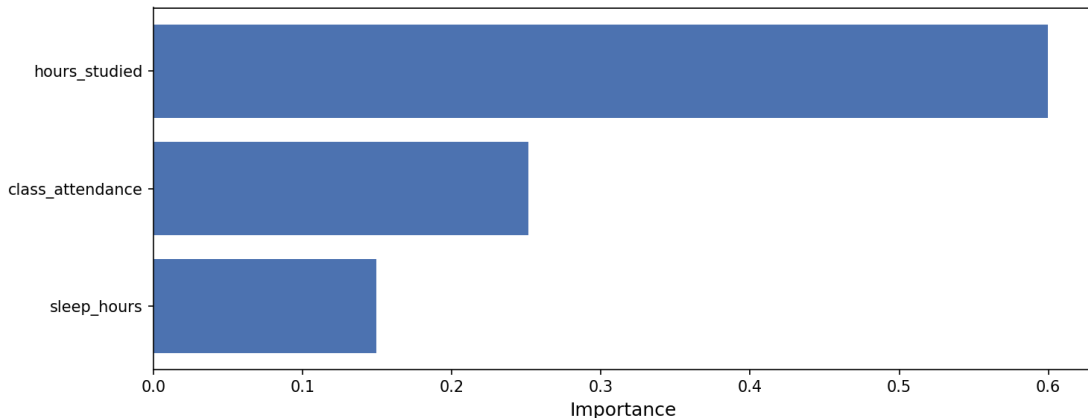
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Random forest: strengths and limitations

Strengths:

- Captures nonlinear relationships and interactions automatically.
- Much more robust than a single tree – far less prone to overfitting.
- Provides feature importance scores for interpretability.
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Limitations:

- Less directly interpretable than a regression equation – no simple β coefficients.
- Slower to train and to predict with than linear/polynomial regression.
- Cannot extrapolate beyond the range of the training data.

Back to our motivating example

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A random forest can capture this kind of **conditional** effect – sleep quality mattering more or less depending on hours studied – **without us having to specify it as an interaction term ourselves.**

Which regression for a given task?

Example task	Model Type	Why?
Predict raw exam score	Linear / Polynomial	Continuous output
Model optimal study hours	Polynomial	Nonlinear relationship
Predict score with complex, unspecified interactions	Random Forest	Captures nonlinearities & interactions automatically